

Asset Financing Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Asset Financing Restraint (AFR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on AFR achieves an annualized gross (net) Sharpe ratio of 0.52 (0.40), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (16) bps/month with a t-statistic of 3.05 (2.32), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets) is 18 bps/month with a t-statistic of 2.57.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately reflect all available information. This departure from the efficient market hypothesis manifests most clearly in the cross-section of stock returns, where numerous characteristics predict future returns with remarkable persistence. While traditional asset pricing models attribute these patterns to risk compensation, an alternative explanation centers on frictions in the information diffusion process that prevent immediate price adjustment.

Among the various channels through which information enters prices, corporate financing decisions represent a particularly rich yet underexplored domain. Firms' choices about asset financing—how they fund their productive investments and manage their capital structure—contain valuable signals about future profitability and growth prospects. However, these signals often emerge gradually through financial statements and require sophisticated analysis to interpret, creating conditions for slow information incorporation. We examine Asset Financing Restraint (AFR), a measure that captures firms' conservative approach to leveraging their asset base, and demonstrate that it predicts cross-sectional returns in a manner consistent with gradual information diffusion rather than risk-based explanations.

The slow diffusion of information provides a compelling framework for understanding why AFR predicts returns. Following the seminal work of [Hong et al. \(2000\)](#) on gradual information flow and [Hirshleifer \(2001\)](#) on limited investor attention, we posit that investors systematically underreact to the information contained in firms' asset financing decisions. When firms exercise restraint in asset financing—maintaining lower leverage relative to their asset base—they signal either superior internal cash generation or disciplined capital allocation. These signals, embedded in financial statements and requiring careful analysis to extract, diffuse slowly through the investor base.

The mechanism operates through multiple channels of informational friction. DellaVigna and Pollet (2009) document that investors exhibit limited attention to information that requires processing complex financial data, while Cohen and Frazzini (2008) show that even sophisticated investors face capacity constraints in processing all available information. Asset financing decisions, which involve interpreting the interaction between balance sheet items and cash flow statements, exemplify the type of complex information that diffuses gradually. Moreover, Peng and Xiong (2007) demonstrate that investors facing attention constraints optimally allocate more attention to market and sector-level information than to firm-specific signals, further slowing the incorporation of firm-level financing restraint measures into prices.

This gradual diffusion framework generates clear empirical predictions. First, we expect return continuation following portfolio formation based on AFR, as prices slowly adjust to reflect the information content. Second, the return predictability should concentrate among stocks with greater informational frictions—smaller firms, those with lower analyst coverage, and limited institutional ownership, as documented by Zhang (2006). Third, the pattern should weaken over time as information eventually diffuses through the market, consistent with the post-earnings announcement drift documented by Bernard and Thomas (1989). These predictions distinguish the gradual diffusion hypothesis from risk-based explanations, which would require AFR to proxy for systematic risk exposure.

We find strong evidence that AFR predicts cross-sectional returns in a pattern consistent with gradual information diffusion. A value-weighted long/short trading strategy based on AFR achieves an annualized gross Sharpe ratio of 0.52 and a monthly average abnormal gross return relative to the Fama-French six-factor model of 22 basis points per month with a t-statistic of 3.05. The strategy’s performance remains robust after accounting for transaction costs, generating an annualized net Sharpe ratio of 0.40 and a monthly net alpha of 16 basis points with a t-statistic of

2.32. These results place AFR among the top decile of documented return predictors in terms of both gross and net performance.

The predictive power of AFR exhibits pronounced variation across the size spectrum in a manner consistent with information-based frictions. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the AFR strategy delivers a monthly return of 36 basis points with a t-statistic of 4.09. The corresponding six-factor alpha equals 28 basis points per month with a t-statistic of 3.11, indicating that the effect persists even among the most liquid and closely followed securities. This concentration of returns among larger stocks, where informational frictions are presumably lower but not absent, suggests that the information in AFR is sufficiently complex to generate underreaction even among relatively efficient market segments.

The strategy’s factor loadings reveal minimal exposure to traditional risk factors, with the most significant loading being 0.28 on the CMA factor with a t-statistic of 5.87. When we control for the six most closely related anomalies from the factor zoo—including Net debt financing, Asset growth, and Change in financial liabilities—the AFR strategy maintains an alpha of 18 basis points per month with a t-statistic of 2.57. This orthogonality to existing predictors suggests that AFR captures a distinct dimension of information that diffuses gradually through markets, rather than simply repackaging known patterns.

Our study makes several contributions to the asset pricing literature on informational frictions and return predictability. We extend the gradual information diffusion framework of [Hong et al. \(2000\)](#) to the domain of corporate financing decisions, demonstrating that complex balance sheet information generates predictable returns even in modern markets with sophisticated information intermediaries. Unlike [Cooper et al. \(2008\)](#), who focus on simple asset growth, we show that the interaction between asset base and financing choices contains incremental information

that markets process slowly. Our evidence complements [Richardson et al. \(2005\)](#) by identifying a specific channel through which accounting information gradually enters prices.

Methodologically, we advance the literature by subjecting AFR to the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), addressing concerns about data mining and spurious patterns that plague the anomalies literature. Our analysis demonstrates that AFR survives multiple robustness tests including alternative portfolio construction methods, double sorts on size, and controls for transaction costs using the [Chen and Velikov \(2022b\)](#) effective spread measures. The strategy generates positive net alphas across all standard factor models, expanding the achievable mean-variance frontier for investors even after accounting for implementation costs. This rigorous approach distinguishes our findings from many documented anomalies that fail to survive such scrutiny.

The broader implications extend beyond documenting another return predictor. Our findings suggest that even fundamental information about corporate financing decisions—data readily available in financial statements—diffuses gradually enough to generate economically significant return predictability. This challenges the notion that technological advances and increased market participation have eliminated informational inefficiencies. The persistence of the AFR effect among large-cap stocks indicates that complexity in information processing, rather than mere availability, drives slow price adjustment. These results support models of bounded rationality and limited attention as essential features of asset pricing, suggesting that informational frictions remain a first-order consideration for understanding cross-sectional return patterns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' long-term debt financing activities. Asset Financing Restraint signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This variable captures new long-term borrowing activities and excludes refinancing of existing debt. Property, plant, and equipment gross (PPEGT) measures the total historical cost of a firm's tangible fixed assets before accumulated depreciation, including land, buildings, machinery, equipment, and construction in progress. This variable provides a measure of the firm's cumulative investment in productive assets. We construct the Asset Financing Restraint signal by calculating the year-over-year change in long-term debt issuance, scaled by lagged gross property, plant, and equipment. Specifically, we compute $(DLTIS(t) - DLTIS(t-1)) / PPEGT(t-1)$ for each firm-year observation. This ratio captures changes in a firm's long-term debt financing relative to its existing asset base, providing insight into shifts in external financing patterns. A decrease in this measure suggests tightening financing constraints or reduced access to debt markets relative to the firm's productive capacity. We calculate the signal using fiscal year-end values and require non-missing values for both DLTIS and PPEGT in consecutive years. To ensure meaningful scaling, we exclude observations where PPEGT equals zero and winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the AFR signal. Panel A plots the time-series of the mean, median, and interquartile range for AFR. On average, the cross-sectional mean (median) AFR is -0.47 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input AFR data. The signal's interquartile range spans -0.18 to 0.18. Panel B of Figure 1 plots the time-series of the coverage of the AFR signal for the CRSP universe. On average, the AFR signal is available for 5.83% of CRSP names, which on average make up 7.10% of total market capitalization.

4 Does AFR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on AFR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high AFR portfolio and sells the low AFR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short AFR strategy earns an average return of 0.26% per month with a t-statistic of 3.73. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.22% to 0.31% per month and have t-statistics exceeding 3.05 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28,

with a t-statistic of 5.87 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 474 stocks and an average market capitalization of at least \$1,675 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 4.02. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective

bid-ask half-spread measure from [Chen and Velikov \(2022a\)](#). The net average returns reported in the first column range between -8-26bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.40. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the AFR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the AFR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and AFR, as well as average returns and alphas for long/short trading AFR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the AFR strategy achieves an average return of 36 bps/month with a t-statistic of 4.09. Among these large cap stocks, the alphas for the AFR strategy relative to the five most common factor models range from 28 to 40 bps/month with t-statistics between 3.11 and 4.55.

5 How does AFR perform relative to the zoo?

Figure 2 puts the performance of AFR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the AFR strategy falls in the distribution. The AFR strategy’s gross (net)

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

Sharpe ratio of 0.52 (0.40) is greater than 91% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the AFR strategy (red line).² Ignoring trading costs, a \$1 invested in the AFR strategy would have yielded \$3.35 which ranks the AFR strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the AFR strategy would have yielded \$2.04 which ranks the AFR strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the AFR relative to those. Panel A shows that the AFR strategy gross alphas fall between the 59 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The AFR strategy has a positive net generalized alpha for five out of the five factor models. In these cases AFR ranks between the 77 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does AFR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of AFR with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price AFR or at least to weaken the power AFR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of AFR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AFR}AFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on AFR. Stocks are finally grouped into five AFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

AFR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on AFR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the AFR signal in these Fama-MacBeth regressions exceed 2.25, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on AFR is 1.32.

Similarly, Table 5 reports results from spanning tests that regress returns to the AFR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the AFR strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.61, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the AFR trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.57.

7 Does AFR add relative to the whole zoo?

Finally, we can ask how much adding AFR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022a\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the AFR signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which AFR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes AFR grows to \$1152.06.

8 Conclusion

This study provides robust evidence that Asset Financing Restraint (AFR) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that AFR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related financing variables.

The key findings reveal that a value-weighted long/short strategy based on AFR delivers an annualized net Sharpe ratio of 0.40, which remains economically significant after accounting for transaction costs. More importantly, the strategy generates a monthly alpha of 16 basis points (t -statistic = 2.32) relative to the Fama-French five-factor model augmented with momentum, indicating that AFR captures unique information about expected returns not subsumed by conventional risk factors. The persistence of a significant alpha (18 bps/month, t -statistic = 2.57) even after controlling for six closely related financing strategies—including net debt financing, net external financing, and asset growth—underscores the distinct predictive content of AFR.

These results have important practical implications for both academic researchers

and investment practitioners. For academics, our findings contribute to the growing literature on the role of financing constraints and capital structure decisions in asset pricing, suggesting that firms' asset financing choices contain valuable information about future returns beyond what is captured by traditional balance sheet growth measures. For practitioners, the AFR signal presents an implementable strategy that maintains economic significance after realistic transaction costs, offering potential for portfolio enhancement.

However, several limitations warrant consideration. First, while our analysis demonstrates statistical and economic significance, the reduction in Sharpe ratio from 0.52 to 0.40 after transaction costs highlights the importance of implementation efficiency. Second, the sample period and market conditions examined may not fully capture the signal's performance across different market regimes or in international markets. Third, the economic mechanisms underlying the AFR anomaly require further theoretical development to fully understand why markets appear to misprice firms based on their asset financing decisions.

Future research should explore several promising avenues. First, investigating the time-varying nature of the AFR premium and its relationship with macroeconomic conditions could provide insights into when the strategy performs best. Second, examining the international evidence for AFR across developed and emerging markets would test the generalizability of our findings. Third, decomposing the sources of AFR's predictive power through detailed analysis of firm characteristics and financing channels could enhance our understanding of the underlying economic mechanisms. Finally, exploring the interaction between AFR and other corporate decisions, such as investment policy and payout choices, may reveal additional insights into how financing constraints affect firm value and returns. Understanding these relationships will be crucial for developing more comprehensive models of how capital structure decisions influence asset prices in equilibrium.

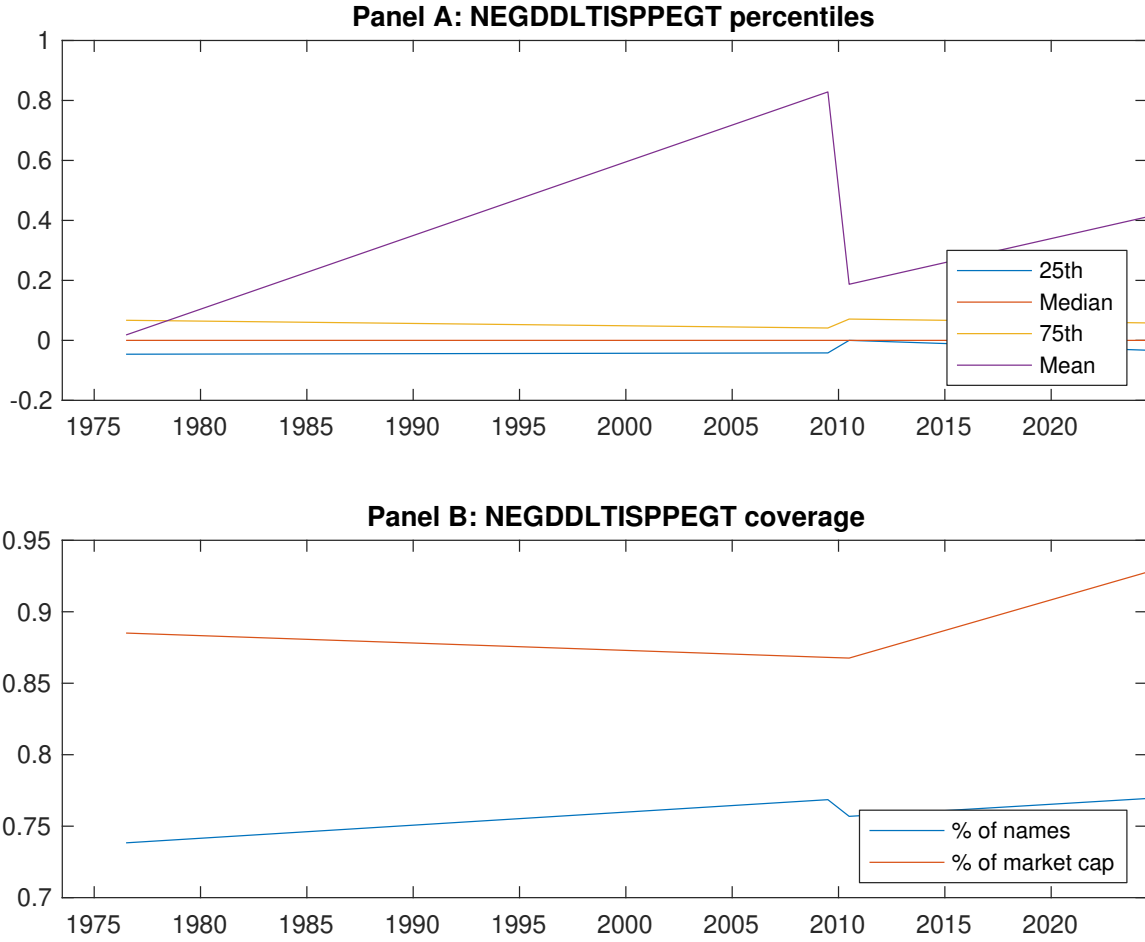


Figure 1: Times series of AFR percentiles and coverage.
This figure plots descriptive statistics for AFR. Panel A shows cross-sectional percentiles of AFR over the sample. Panel B plots the monthly coverage of AFR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on AFR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on AFR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.62 [2.84]	0.66 [3.77]	0.62 [3.22]	0.70 [4.01]	0.88 [4.30]	0.26 [3.73]
α_{CAPM}	-0.13 [-2.29]	0.07 [1.35]	-0.03 [-0.46]	0.11 [2.21]	0.18 [3.43]	0.31 [4.44]
α_{FF3}	-0.12 [-2.14]	0.05 [1.02]	0.03 [0.57]	0.11 [2.24]	0.20 [3.81]	0.31 [4.46]
α_{FF4}	-0.09 [-1.63]	0.05 [1.08]	0.08 [1.42]	0.07 [1.36]	0.18 [3.50]	0.27 [3.86]
α_{FF5}	-0.09 [-1.74]	-0.04 [-0.85]	0.06 [1.03]	0.03 [0.61]	0.14 [2.74]	0.24 [3.38]
α_{FF6}	-0.08 [-1.42]	-0.03 [-0.63]	0.09 [1.67]	0.00 [0.05]	0.14 [2.62]	0.22 [3.05]
Panel B: Fama and French (2018) 6-factor model loadings for AFR-sorted portfolios						
β_{MKT}	1.10 [86.47]	0.97 [86.31]	0.96 [73.15]	0.95 [82.62]	1.05 [86.53]	-0.05 [-2.80]
β_{SMB}	0.10 [5.18]	-0.11 [-6.52]	-0.02 [-0.82]	-0.05 [-2.66]	0.15 [8.16]	0.05 [2.05]
β_{HML}	0.00 [0.00]	0.02 [0.99]	-0.16 [-6.28]	-0.05 [-2.32]	-0.13 [-5.56]	-0.13 [-4.14]
β_{RMW}	0.05 [2.16]	0.15 [6.81]	-0.02 [-0.73]	0.09 [3.91]	0.07 [2.88]	0.01 [0.47]
β_{CMA}	-0.15 [-4.06]	0.15 [4.67]	-0.05 [-1.23]	0.18 [5.34]	0.13 [3.65]	0.28 [5.87]
β_{UMD}	-0.03 [-2.10]	-0.02 [-1.47]	-0.06 [-4.37]	0.04 [3.89]	0.01 [0.63]	0.03 [2.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	668	474	995	516	644	
me (\$10 ⁶)	1675	2634	2004	2612	1678	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the AFR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.73]	0.31 [4.44]	0.31 [4.46]	0.27 [3.86]	0.24 [3.38]	0.22 [3.05]
Quintile	NYSE	EW	0.18 [4.02]	0.20 [4.65]	0.19 [4.35]	0.18 [4.16]	0.18 [4.06]	0.18 [4.00]
Quintile	Name	VW	0.27 [3.86]	0.31 [4.40]	0.32 [4.49]	0.29 [4.09]	0.25 [3.57]	0.24 [3.41]
Quintile	Cap	VW	0.26 [4.00]	0.31 [4.78]	0.31 [4.86]	0.27 [4.22]	0.23 [3.57]	0.21 [3.23]
Decile	NYSE	VW	0.33 [3.52]	0.39 [4.11]	0.37 [3.89]	0.34 [3.52]	0.23 [2.42]	0.22 [2.32]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.20 [2.88]	0.24 [3.46]	0.24 [3.45]	0.22 [3.14]	0.18 [2.56]	0.16 [2.32]
Quintile	NYSE	EW	-0.08 [-1.40]					
Quintile	Name	VW	0.21 [2.98]	0.24 [3.40]	0.24 [3.44]	0.23 [3.25]	0.19 [2.71]	0.18 [2.55]
Quintile	Cap	VW	0.21 [3.18]	0.25 [3.96]	0.26 [4.00]	0.24 [3.68]	0.19 [2.93]	0.17 [2.69]
Decile	NYSE	VW	0.26 [2.74]	0.32 [3.34]	0.30 [3.12]	0.28 [2.95]	0.18 [1.91]	0.17 [1.80]

Table 3: Conditional sort on size and AFR

This table presents results for conditional double sorts on size and AFR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on AFR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high AFR and short stocks with low AFR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	AFR Quintiles					AFR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.67 [2.28]	0.87 [3.29]	0.97 [3.50]	0.92 [3.40]	0.84 [2.78]	0.16 [1.61]	0.20 [1.97]	0.17 [1.69]	0.14 [1.30]	0.13 [1.25]	0.11 [1.04]
	(2)	0.75 [2.75]	0.88 [3.55]	0.85 [3.34]	0.91 [3.74]	0.85 [3.29]	0.10 [1.18]	0.13 [1.65]	0.10 [1.32]	0.12 [1.50]	0.06 [0.80]	0.08 [1.00]
	(3)	0.83 [3.22]	0.78 [3.56]	0.83 [3.47]	0.80 [3.66]	0.94 [3.88]	0.11 [1.43]	0.16 [1.98]	0.15 [1.93]	0.11 [1.39]	0.16 [1.95]	0.13 [1.56]
	(4)	0.76 [3.25]	0.81 [3.85]	0.82 [3.71]	0.77 [3.77]	0.87 [3.92]	0.11 [1.49]	0.15 [1.94]	0.13 [1.66]	0.10 [1.25]	0.10 [1.31]	0.08 [1.04]
	(5)	0.53 [2.60]	0.63 [3.56]	0.58 [3.12]	0.58 [3.24]	0.89 [4.57]	0.36 [4.09]	0.40 [4.48]	0.40 [4.55]	0.34 [3.79]	0.32 [3.58]	0.28 [3.11]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	AFR Quintiles					AFR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	367	368	369	368	364	33	30	29	29	32	
	(2)	101	101	101	102	101	55	56	55	56	55	
	(3)	72	73	72	73	73	100	100	96	97	100	
	(4)	61	62	62	62	61	219	225	218	222	217	
(5)	57	57	57	57	57	1394	2002	1667	2045	1472		

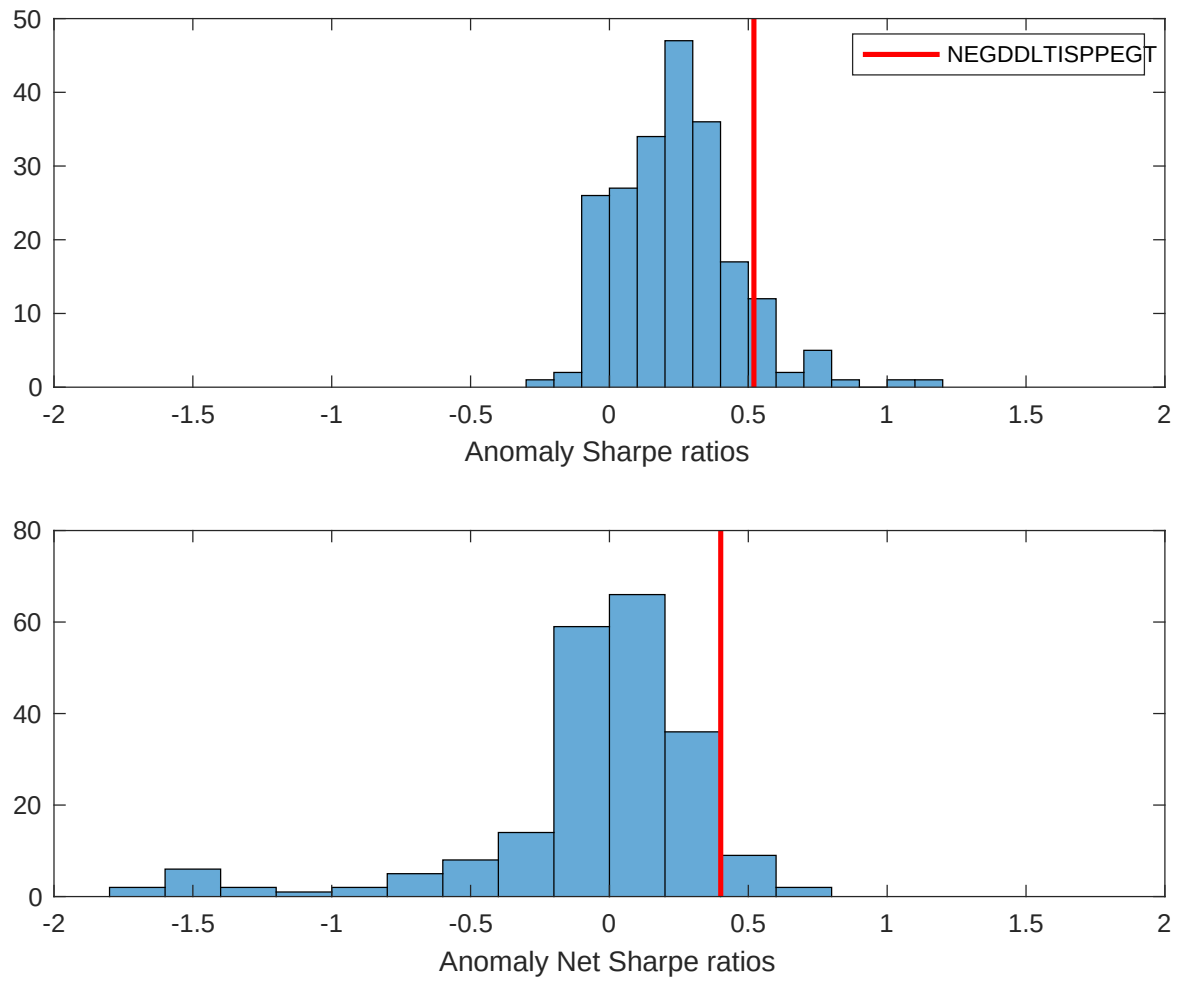


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the AFR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

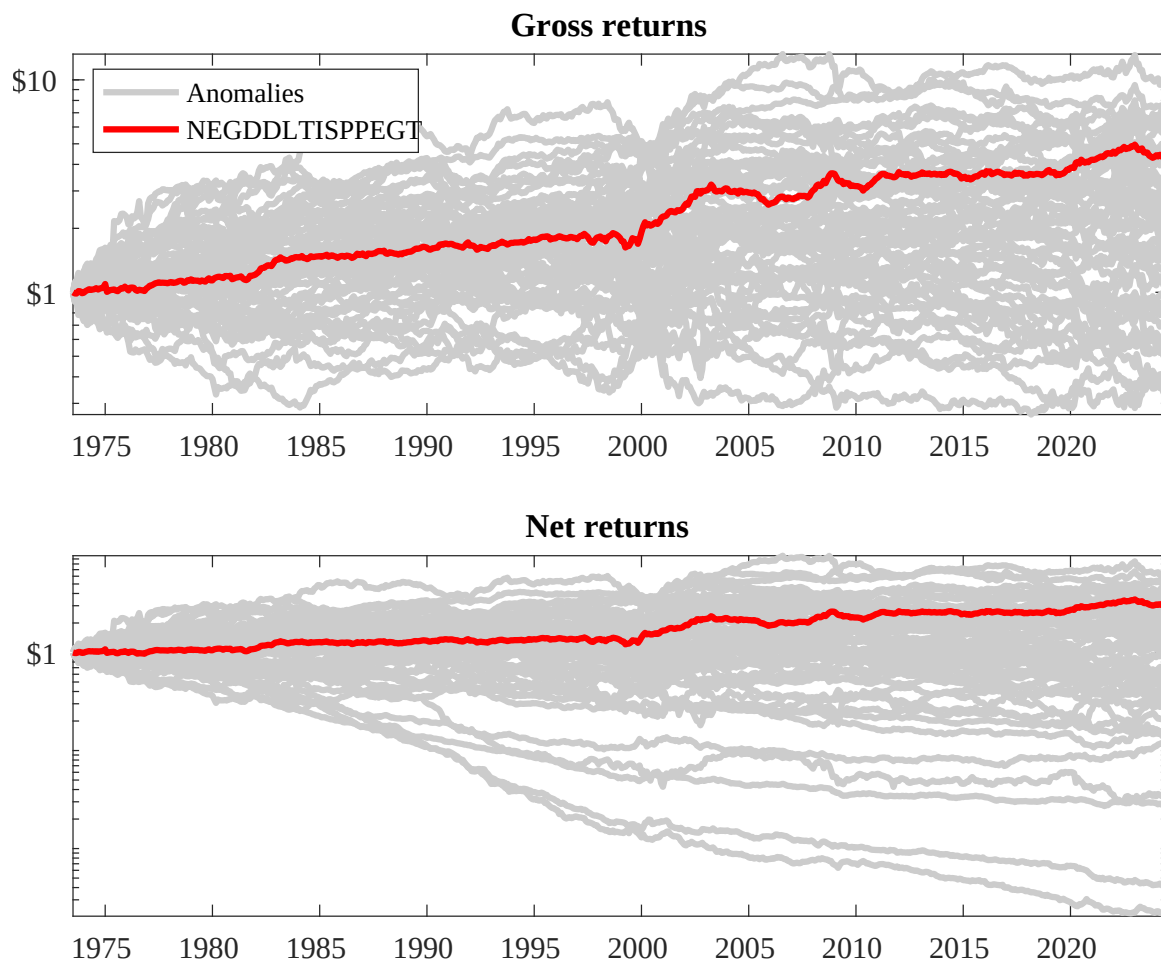
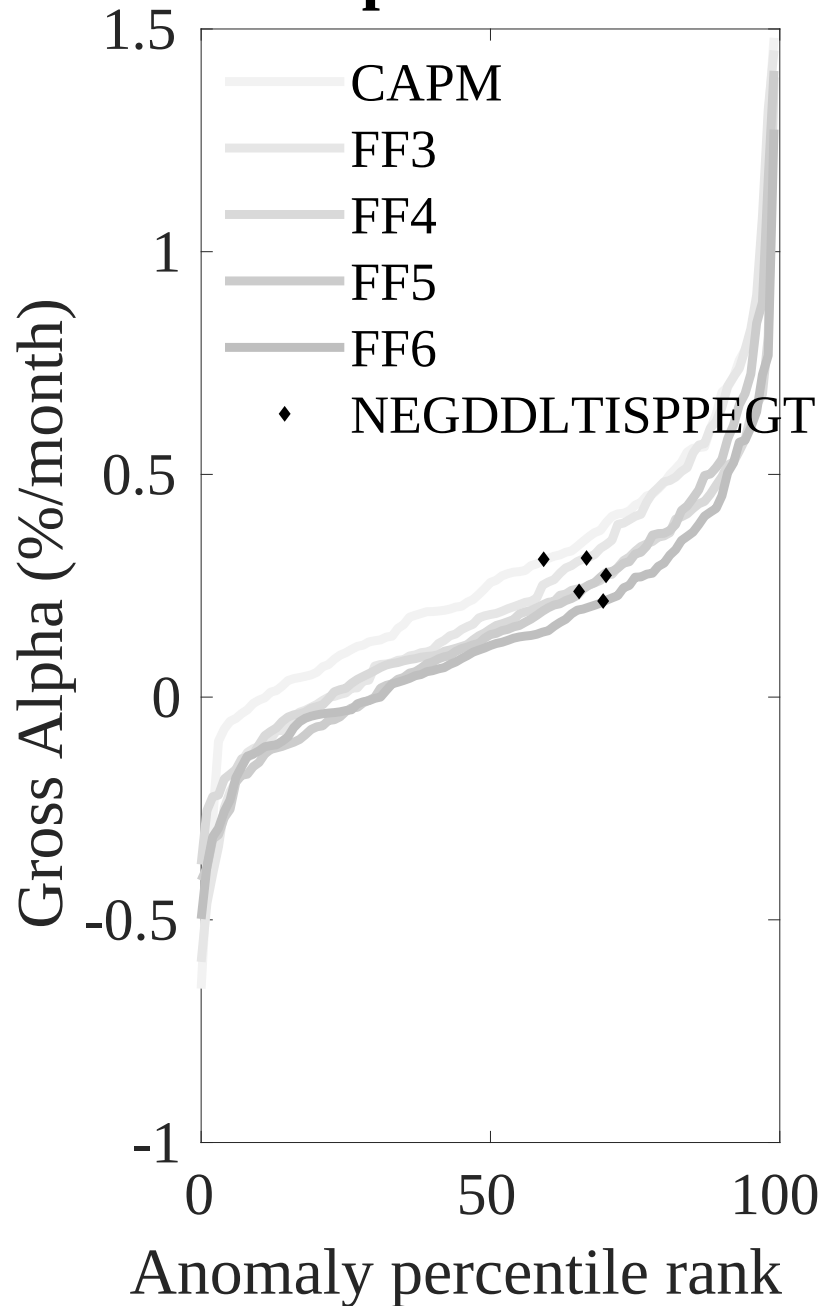


Figure 3: Dollar invested.

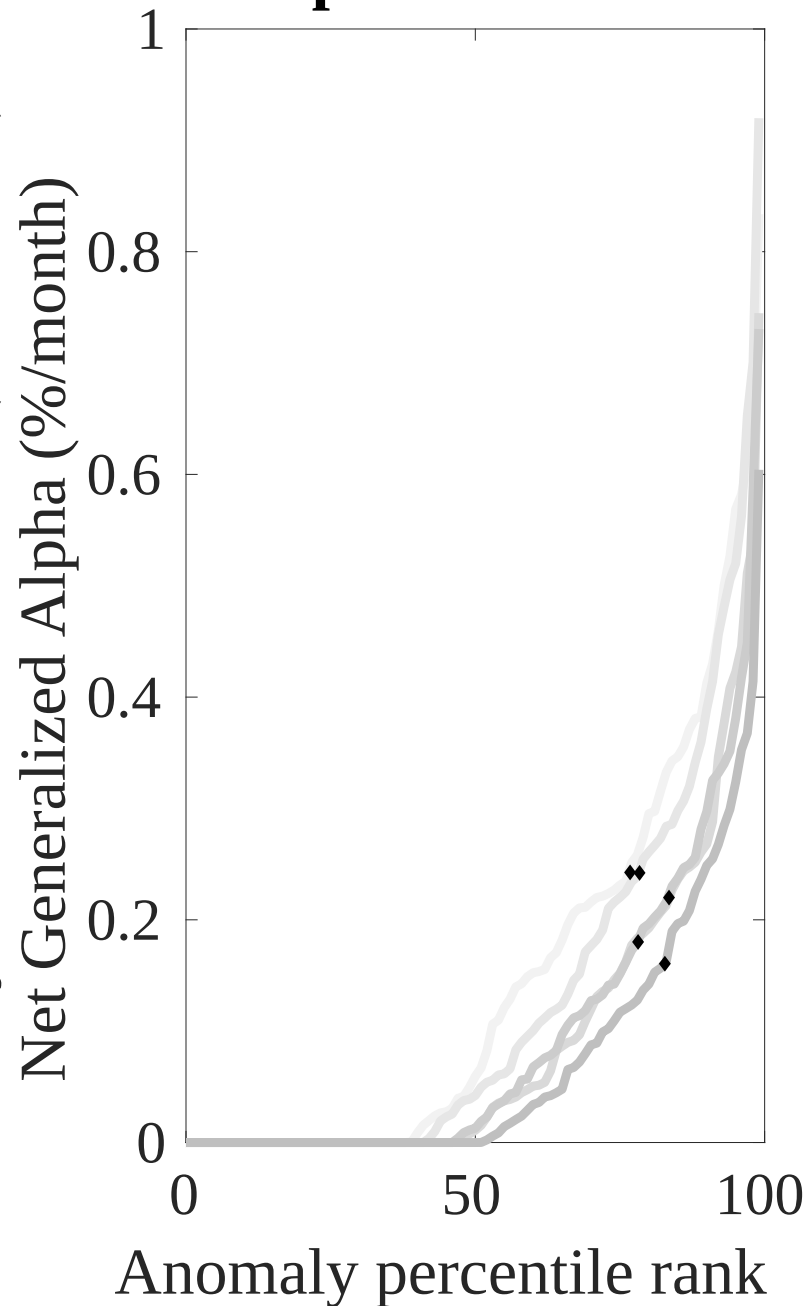
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the AFR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



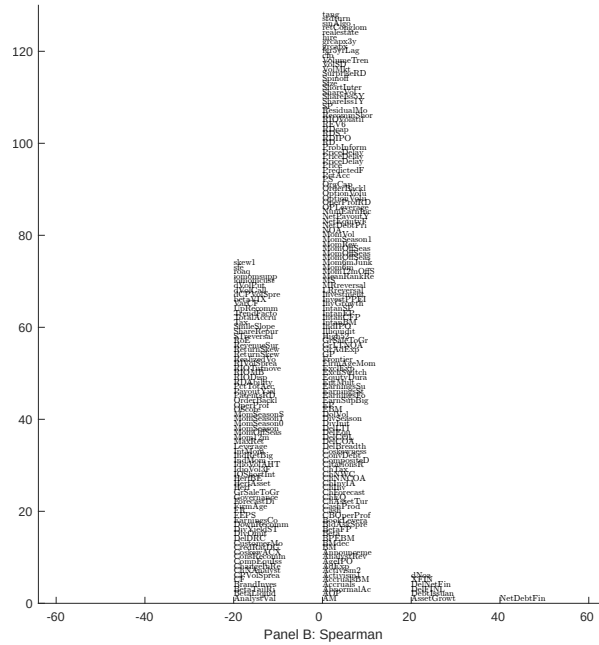
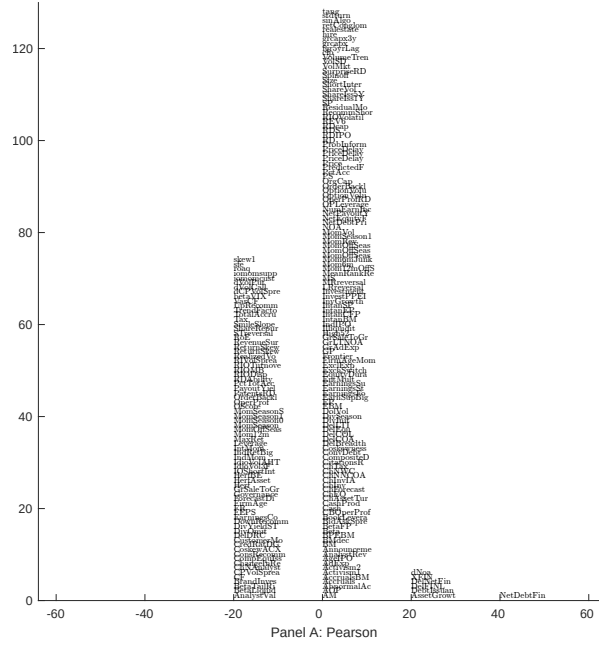


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with AFR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

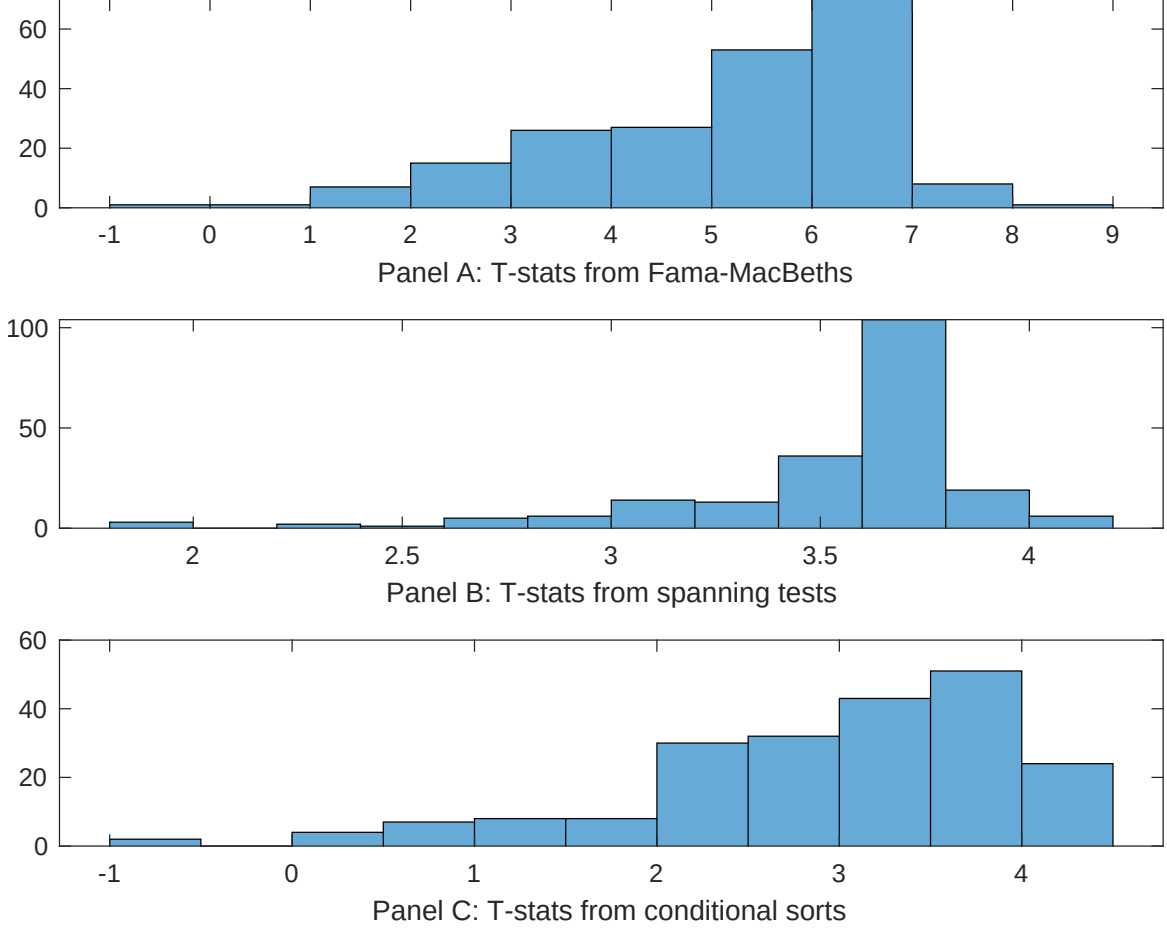


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of AFR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AFR} AFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on AFR. Stocks are finally grouped into five AFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AFR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on AFR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{AFR}AFR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.35]	0.14 [5.72]	0.13 [5.38]	0.14 [5.79]	0.14 [5.31]	0.14 [5.76]	0.15 [5.75]
AFR	0.88 [3.06]	0.10 [4.00]	0.75 [2.94]	0.64 [2.25]	0.15 [5.11]	0.95 [3.51]	0.42 [1.32]
Anomaly 1	0.19 [8.53]						0.91 [1.34]
Anomaly 2		0.19 [6.30]					0.11 [1.82]
Anomaly 3			0.17 [9.16]				-0.79 [-1.65]
Anomaly 4				0.10 [8.27]			0.51 [2.35]
Anomaly 5					0.41 [6.94]		0.25 [0.42]
Anomaly 6						0.17 [7.59]	0.86 [3.10]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the AFR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{AFR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Net external financing, Change in financial liabilities, Asset growth, Inventory Growth, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.73]	0.18 [2.61]	0.19 [2.74]	0.20 [2.85]	0.20 [2.93]	0.20 [2.89]	0.18 [2.57]
Anomaly 1	15.76 [4.15]						11.45 [2.21]
Anomaly 2		17.80 [5.16]					14.30 [3.77]
Anomaly 3			9.47 [2.39]				-0.24 [-0.05]
Anomaly 4				9.79 [2.41]			3.87 [0.92]
Anomaly 5					4.97 [1.82]		3.92 [1.37]
Anomaly 6						4.40 [1.39]	-1.83 [-0.55]
mkt	-4.23 [-2.65]	-1.81 [-1.09]	-4.04 [-2.51]	-4.26 [-2.64]	-4.35 [-2.70]	-4.32 [-2.67]	-2.30 [-1.38]
smb	4.04 [1.66]	11.15 [4.22]	4.46 [1.81]	4.72 [1.92]	5.98 [2.43]	5.45 [2.23]	9.17 [3.26]
hml	-11.92 [-3.87]	-10.59 [-3.44]	-11.92 [-3.83]	-13.02 [-4.18]	-12.71 [-4.08]	-13.02 [-4.15]	-10.67 [-3.42]
rmw	-0.10 [-0.03]	-9.01 [-2.42]	0.78 [0.25]	1.67 [0.53]	2.35 [0.74]	1.71 [0.54]	-7.61 [-1.99]
cma	24.20 [5.07]	16.69 [3.24]	25.11 [5.12]	16.44 [2.42]	23.86 [4.49]	24.79 [4.63]	9.40 [1.34]
umd	2.63 [1.64]	3.52 [2.24]	2.83 [1.73]	3.85 [2.40]	3.14 [1.93]	3.64 [2.27]	2.50 [1.53]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	14	11	11	11	10	15

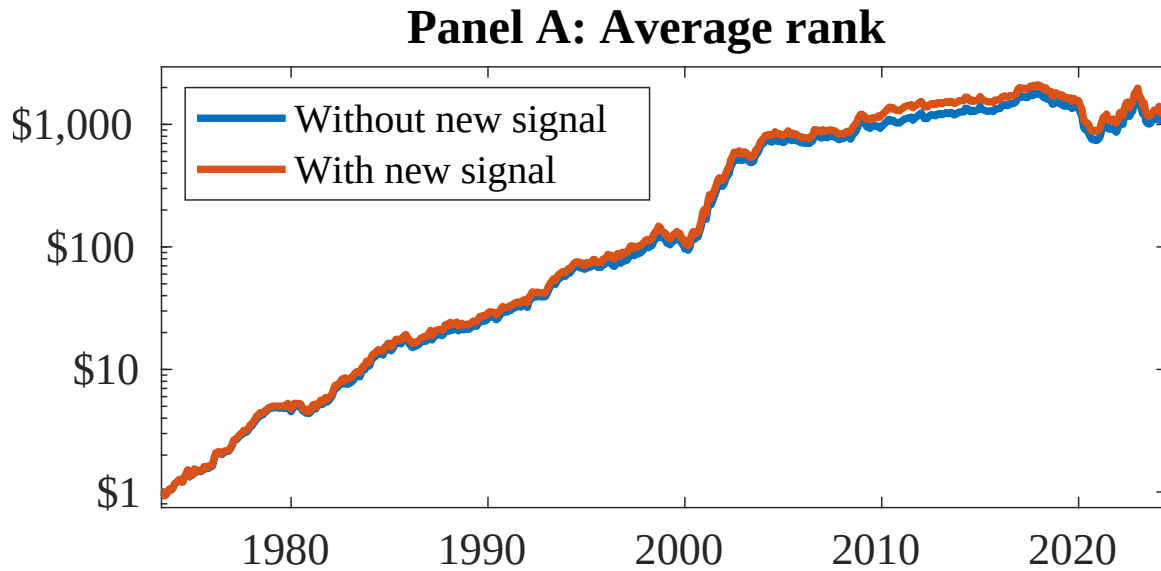


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022a\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as AFR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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